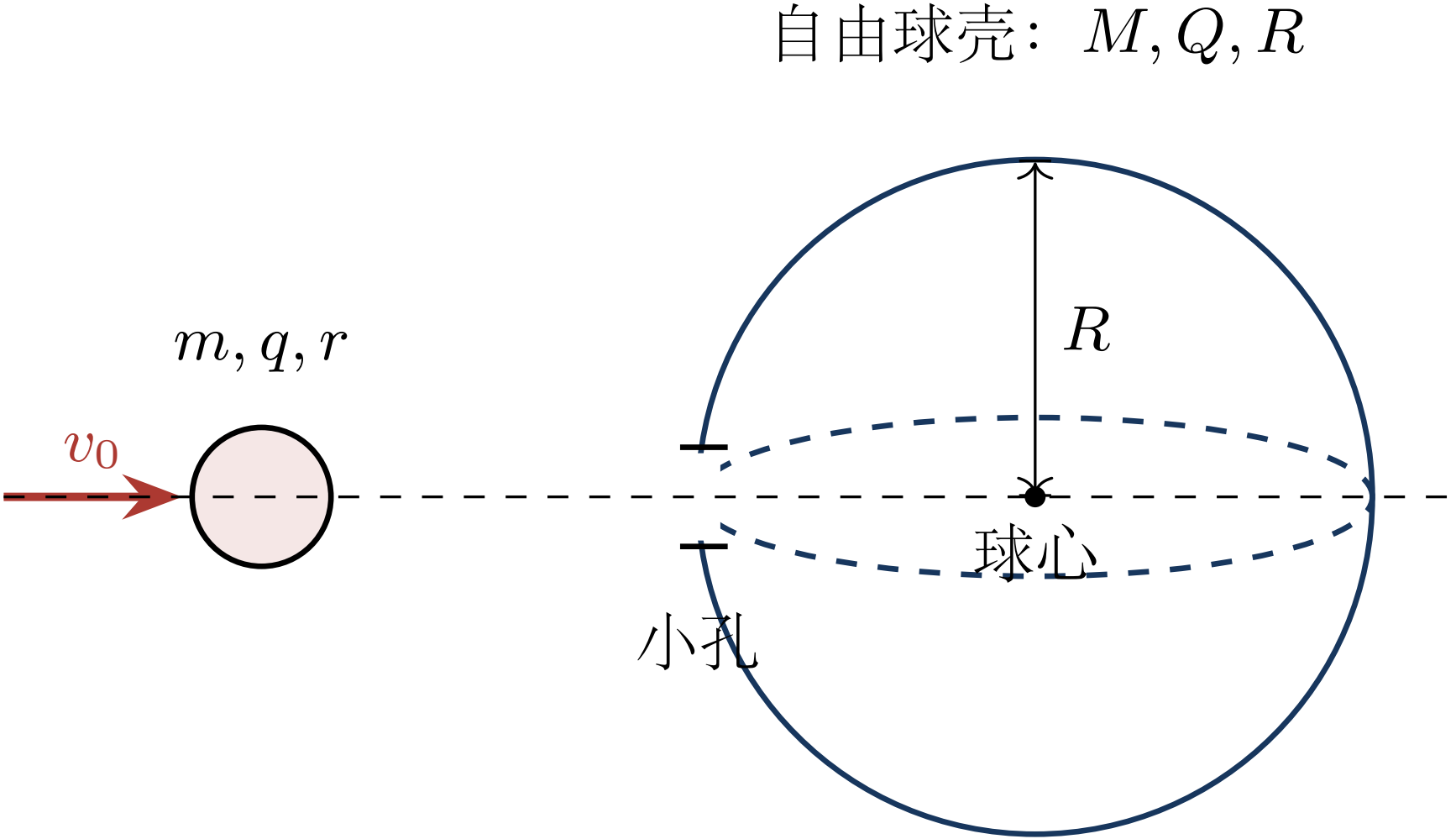
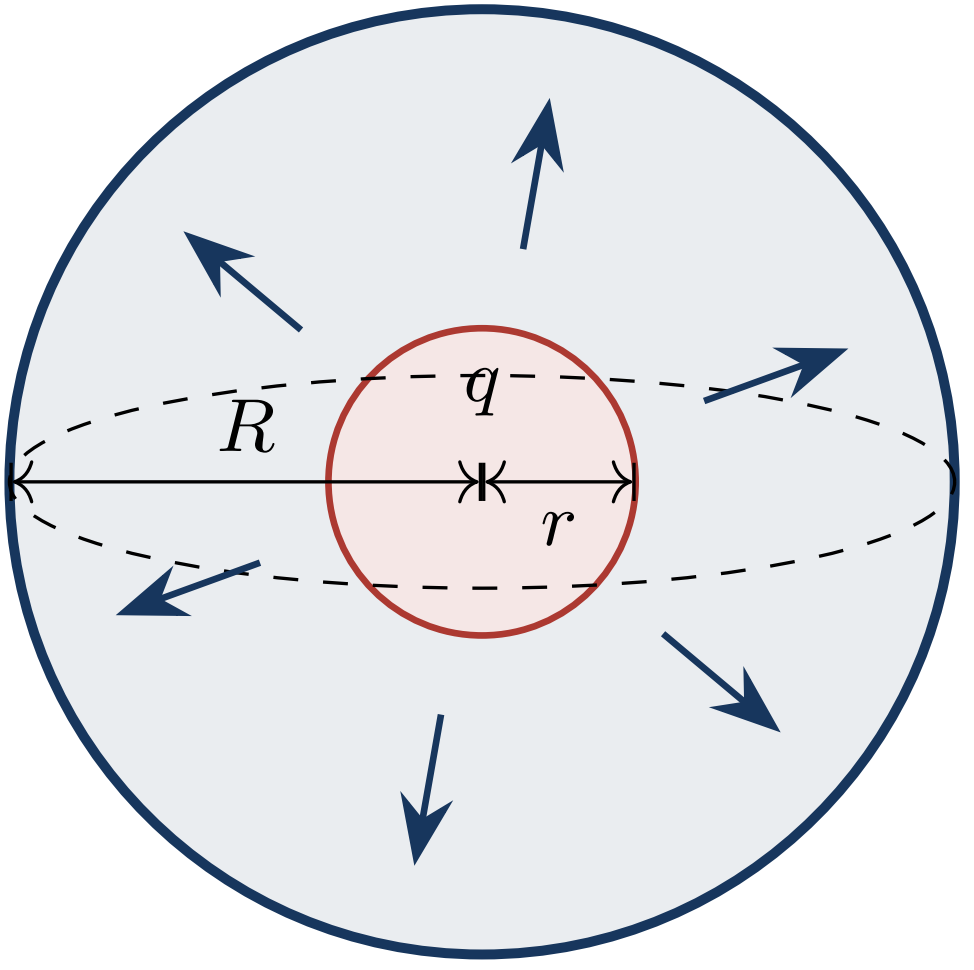

自由球壳：M, Q, R
m, q, r
v_0
R
球心
小孔

内壁 $-q$，外表面 $Q+q$



$r < s < R$：

$E = kq/s^2$

$s > R$：

$E = k(Q+q)/s^2$

金属中 $E = 0$，两区域场能分别积分

质心 $V_c = mv_0/(m+M)$

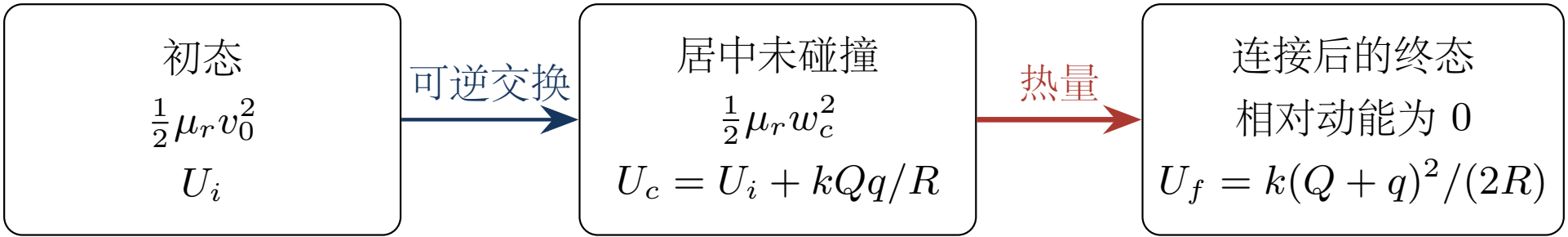
小球

$u = V_c + Mw/(m+M)$

球壳

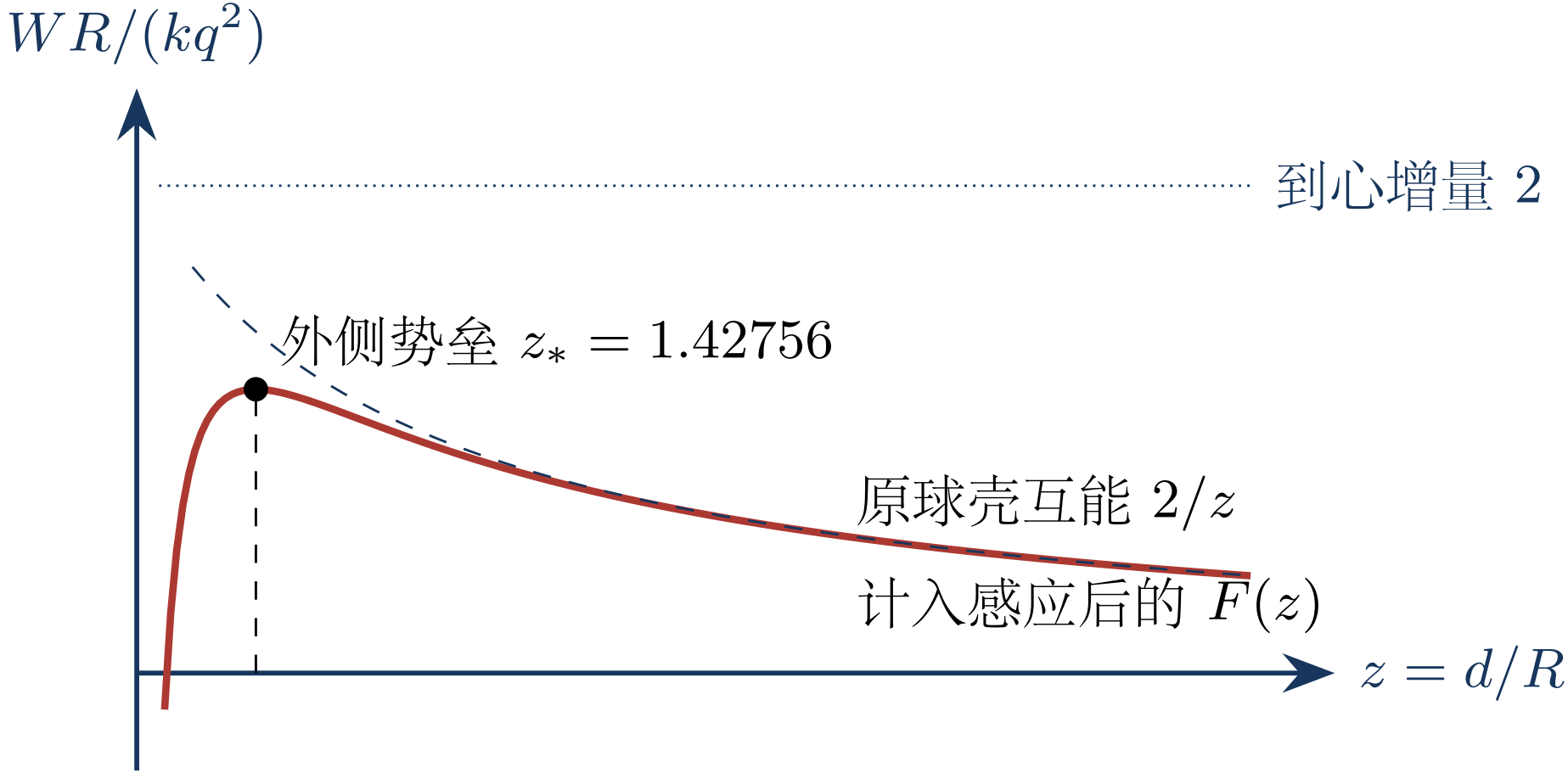
$V = V_c - mw/(m+M)$

动量不变，相对动能变化：$\mu_r = mM/(m+M)$



全过程质心动能保持：$\frac{1}{2}(m+M)V_c^2$

$$Q_{热} = \frac{1}{2}\mu_r w_c^2 + \frac{kq^2}{2}(1/r - 1/R)$$



示例 $Q/q = 2$；进入门槛低于居中能量条件